Exploratory modelling of collective effects in the FCC-ee High-Energy Booster using machine learning

Master's thesis defense presentation

Santiago Martinez Saenz[1]

[1]GANIL/CNRS – Université Paris-Saclay / Università di Roma La Sapienza

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Preliminary results published in the JACoW proceedings of IPAC 2026

The presentation separates established physics from the work developed in this work

Scientific context

1. FCC-ee HEB and collective instabilities.
2. Distribution-level beam dynamics.
3. Simulation cost and prior surrogate methods.

Main contributions

1. Reproducible Xsuite/PyColleff datasets and projected 6D representation.
2. FNO, CVAE, Transformer, local-window Transformer and GNO surrogates.
3. Common endpoint comparison: distributions, physical statistics and runtime.
4. Conditional latent-uncertainty study, limitations and next validation steps.

Sources: Preliminary FCC-ee HEB results: (Ghribi et al., 2026b).

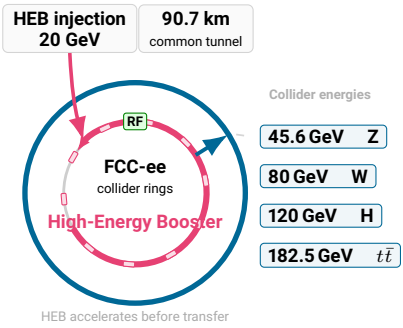
The FCC-ee HEB must preserve beam quality while ramping through a 90.7 km accelerator complex

Role in the injection chain

- FCC-ee is the leading flagship project in CERN's post-LHC programme.
- The HEB follows the high-energy injector stage and shares the collider-tunnel infrastructure.
- It ramps from about 20 GeV to the collider injection energies 45.6–182.5 GeV.
- Its task is to deliver the required intensity and phase-space quality for top-up injection.

Machine complexity

The exact lattice-element count is not fixed. The HEB full lattice in the present work contains about 17,000 elements. The lattice is a simplified version of the FCC-ee HEB lattice v24.1.



Sources: (FCC Collaboration, 2025; FCC-ee High-Energy Booster Optics Team, 2026; Ghribi et al., 2026a; Zimmermann et al., 2024).

Collective forces make beam transport a nonlinear operator on the distribution

Beam-distribution evolution

$$\rho_s(\xi) \longmapsto \rho_{s+\Delta s}(\xi), \quad \xi = (x, p_x, y, p_y, \zeta, \delta).$$

The force is separated into physical pieces:

$$\frac{\partial \rho_s}{\partial s} + \nabla_\xi \cdot [\rho_s (F_{\text{ext}} + F_{\text{coll}}[\rho_s] + F_{\text{rad}})] = \frac{1}{2} \nabla_\xi \nabla_\xi : (D_{\text{rad}} \rho_s).$$

Operator-learning target

The validated object is the endpoint map

$$\mathcal{G}_\mu : \rho_{\text{in}} \longrightarrow \rho_{\text{out}}.$$

Sources: (Métral, 2021; Wiedemann, 2015).

First-order operator splitting used as the conceptual model

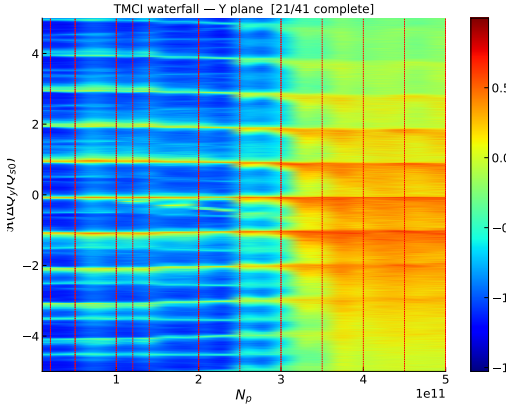
$$\rho_{j+1} \approx e^{\Delta s \mathcal{D}_j} e^{\Delta s \mathcal{C}_j[\rho_j]} (M_{j,\mu_j})_{\#} \rho_j.$$

external lattice → collective kick → radiation / diffusion

Key learning objective

The model is focused in the nonlinear collective operator, which is the only piece that depends on the distribution itself.

High-particle tracking is cheap once, but repeated scans turn it into a CPU-budget problem



Timing retained

Stage	t_{sample}	CPU-core h	
		2048	10^5
Xsuite, 10^6 particles	1.41 s	0.807	39.2
PyColleff, $N_\zeta = 2048$	2.72 s	1.547	75.6
6D histograms	2.32 ms	1.32×10^{-3}	6.44×10^{-2}

CC-IN2P3 `gpu_v100` host: $4\times$ NVIDIA Tesla V100 PCIe (32 GB HBM2), 192 GB RAM and two Intel Xeon Silver 4114/4214R CPUs, 20–24 cores per node. One fully allocated node-hour therefore represents 20–24 CPU-core h.

Simulation record required for this scan

The data has been simulated for 1000 turns, 20 GeV, using the full longitudinal wake/impedance budget, Z mode. IBS, space charge and synchrotron radiation disabled. All has been simulated at CC-IN2P3 in CPU mode.

Sources: (Iadarola et al., 2023; Xsuite Collaboration, 2026; CC-IN2P3, 2026; Rigaud, 2021; FCC-ee High-Energy Booster Optics Team, 2026).

The 6D learning set contains 2048 paired full-lattice Xsuite trajectories with fixed splits

Reproducible generation chain

FCC-ee HEB MAD-X optics → cpymad/Xtrack → PyWIT wake/impedance → $(Z_{\mathrm{in}}, \mu, Z_{\mathrm{out}})$.

$$\mathcal{D}_{\mathrm{track}} = \left\{ \left(Z_{\mathrm{in}}^{(s)}, \mu^{(s)}, Z_{\mathrm{out}}^{(s)} \right) \right\}_{s=1}^{2048}.$$

10^6 particles in the recorded timing, and retains 4096 macroparticles in each stored cloud.

Scope

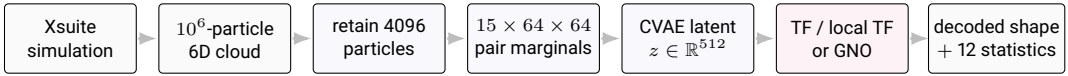
The dataset is a fixed archive of 2048 paired trajectories. The training, validation and test splits are also fixed.

Sources: (FCC-ee High-Energy Booster Optics Team, 2026; Iadarola et al., 2023; ImpedanCEI Collaboration, 2026; Martinez Saenz, 2026a,b).

Fixed archive

Total	2048
Train	1640
Validation	204
Test	204
Retained particles/cloud	4096

The pipeline separates sampled 6D shape from the physical position and scale of the beam



Compression and representation

A direct 64^6 density grid is infeasible. Fifteen pairwise marginals retain measurable non-Gaussian cores, tails and two-coordinate correlations while allowing convolutional compression.

Phase-space tomography

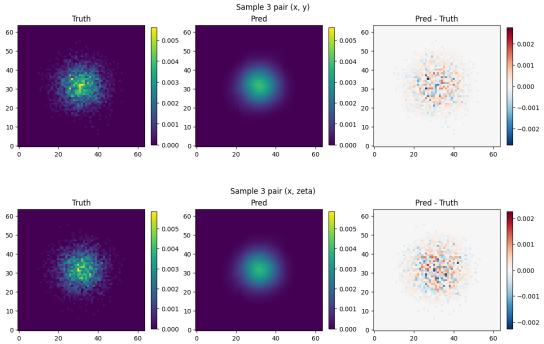
Low-dimensional projections are the observables from which high-dimensional beam distributions are inferred. As such the decompression is not unique, but can be trained to preserve the measured marginals and the known physical statistics.

Sources: (Goutierre et al., 2023; Hoover and Wong, 2024; Roussel et al., 2023).

The CVAE compresses projected shape while conditioning keeps known beam information explicit

Conditional statistics match the known beam information

- Direct propagation of $15 \times 64 \times 64 = 61,440$ bins is expensive.
- A stochastic latent state gives a continuous, sampleable representation of projected shape.
- Explicit conditioning retains known statistics and domain variables, posterior sampling enables the later sensitivity study.

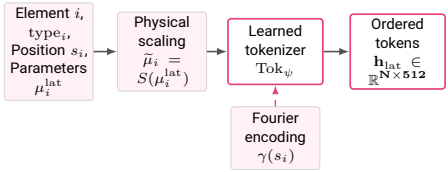


Sources: (Kingma and Welling, 2014; Sohn et al., 2015).

The CVAE is a stochastic encoder-decoder that learns a posterior distribution over a latent space

Probabilistic motivation for the CVAE

- A deterministic autoencoder predicts a single realization, the CVAE learns $q_\phi(z \mid H, c, \mu)$.
- Conditioning $c = [\sigma, m]$ keeps known scale, centroid and domain information explicit.
- The sampleable 512-state can be propagated by Transformer or neural operators and later resampled for latent-sensitivity analysis.



Implemented posterior and objective

$$q_\phi(z \mid H, c, \mu) = \mathcal{N}(m_z[\mu], \operatorname{diag} \sigma_z^2),$$
$$z = m_z[\mu] + \sigma_z \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The prior is fixed, the encoder learns m_z and σ_z^2. Reparameterization keeps sampling differentiable.

Sources: (Wang et al., 2026; Kingma and Welling, 2014; Sohn et al., 2015).

Lattice tokenization

The CVAE compresses beam shape to z_0, while the tokenizer retains lattice order and local physical descriptors through $\{h_i\}_{i=1}^N$.

Attention might not be All You Need , but it is a powerful tool for sequence modeling

Attention mechanism

Each lattice element has a token h_j. For the update at i:

$$q_i = W_Q h_i, \quad k_j = W_K h_j, \quad v_j = W_V h_j.$$

What the current element needs; how upstream element j is matched; information that element j can contribute.

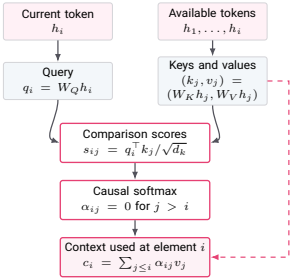
Compare, normalize, then combine

$$s_{ij} = \frac{q_i^\top k_j}{\sqrt{d_k}}, \quad j \le i,$$

$$\alpha_{ij} = \mathrm{softmax}_{j \le i}(s_{ij}), \quad \sum_{j \le i} \alpha_{ij} = 1,$$

$$c_i = \sum_{j \le i} \alpha_{ij} v_j.$$

Larger α_{ij} means a stronger contribution from element j.



Global causal attention

Any $h_1, \ldots, h_i$ may contribute; downstream tokens are masked.

Sources: Original Transformer formulation: (Vaswani et al., 2017).

Global causal attention respects lattice order but permits dense upstream mixing

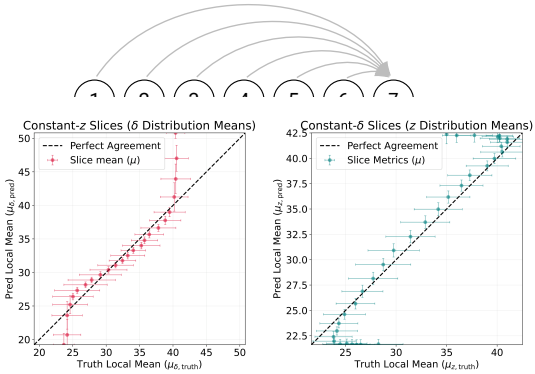
Element-conditioned latent sequence

$$z_{i+1} = z_i + \Delta z_i, \quad \Delta z_i = f_\theta(z_i, h_i, \mu).$$

Causal masking enforces upstream → downstream order, but token i can directly aggregate all $j \leq i$.

Selected endpoint checkpoint

$$\epsilon_{\mathrm{rel}} = 0.681, \quad \mathrm{SW} = 0.087,$$
$$\tau = 18.74\ \mathrm{ms/sample}.$$



Conditional-slice diagnostic: coherent longitudinal distortion is visible, but the figure does not prove global attention is its unique cause.

Sources: (Vaswani et al., 2017; Wang et al., 2026).

Restricting direct attention support improves the endpoint task without changing the representation

Local causal support

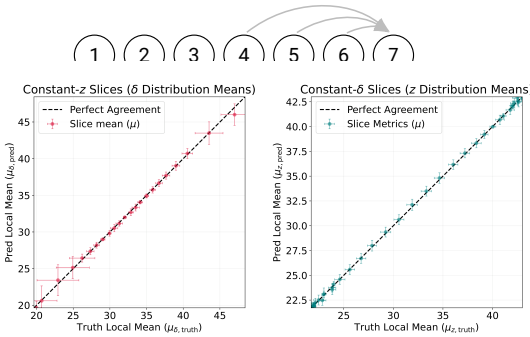
$$M_{ij}^{(w)} = \begin{cases} 0, & 0 \le i - j \le w, \\ -\infty, & \text{otherwise.} \end{cases}$$

Stacked layers still build longer-range dependence through repeated local composition.

Held-out endpoint change at $w = 8$

$$\epsilon_{\text{rel}} : 0.681 \to 0.534,$$
$$\text{SW} : 0.087 \to 0.022.$$

Only the admissible support of direct information transfer is changed.



Local-window Transformer conditional-slice result for the longitudinal (ζ, δ) marginal.

Sources: (Vaswani et al., 2017).

Neural operators match the mathematical type of distribution evolution

Finite-dimensional network

$$x \in \mathbb{R}^n \longmapsto y \in \mathbb{R}^m.$$

The map is tied to finite dimensional vector representations of the data.

Operator model

$$a(\cdot) \longmapsto u(\cdot), \qquad \rho_{\text{in}} \longmapsto \rho_{\text{out}}.$$

The object transformed is a field, profile or distribution.

Generic operator layer

$$v_{\ell+1}(x) = \sigma\Big(\underbrace{W_\ell v_\ell(x)}_{\text{local update}} + \int_D \underbrace{\kappa_\theta(x, y; \mu)}_{\text{learned interaction kernel}} \quad v_\ell(y)\, \mathrm{d}\nu(y) \Big).$$

- The architecture chooses how to parameterize the kernel κ_θ and which points are allowed to interact.
- FNO and GNO are two different kernel parameterizations of this same operator-learning idea.

Sources: (Kovachki et al., 2021, 2023; Li et al., 2020).

FNO is a controlled longitudinal benchmark because wake response is already spectral

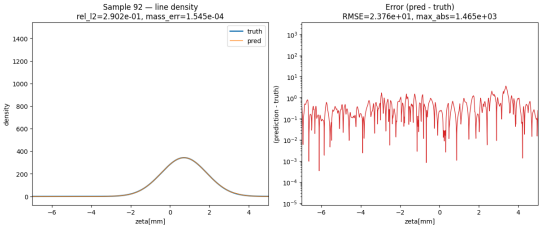
Physical and learned operators use the same representation

$$V_{\mathrm{ind}}(z) = -\mathcal{F}^{-1}[Z_{\parallel}(\omega;\mu)\widehat{\lambda}(\omega)](z),$$
$$\mathcal{K}_{\theta}^{\mathrm{FNO}} v = \mathcal{F}^{-1}[R_{\theta}(k)\mathcal{F}(v)(k)].$$

Benchmark scope

2048 PyColleff samples, 2048 longitudinal grid points, 3 GeV ring, $Q = 10^4$, three relaxation steps with $\alpha = 0.7$. This is separate from the 6D HEB dataset.

Sources: (Kovachki et al., 2023; de Sá and Alves, 2024; Zhou et al., 2023).



Representative line-density prediction and residual. Qualitative validation on the FNO by working in reduced dimension.

A GNO represents lattice propagation as a learned kernel on a directed graph

Operator origin

$$\mathcal{K}_\theta[v](x) = \int_D \kappa_\theta(x, y, \mu) v(y)\, \mathrm{d}\nu(y)$$

becomes, on graph nodes,

$$\mathcal{K}_\theta[v](x_i) \approx \sum_{j \in \mathcal{N}(i)} \omega_{ij} \kappa_\theta(x_i, x_j, \mu) v_j.$$

GNOs were introduced as graph-kernel neural operators for PDE solution maps and irregular domains.

GNO appears architecturally well-suited to accelerator propagation

- Accelerator elements form an ordered, nonuniform sequence,
- Directed edges enforce admissible upstream information transfer,
- The learned kernel can use element type, position, relative geometry and global parameters,
- Repeated local messages mirror composition of element-conditioned transformations.

Prior accelerator latent-dynamics work conditions learned propagation on element descriptors. The present work tests a graph-kernel implementation for the FCC-ee HEB endpoint map.

Sources: (Kovachki et al., 2023; Li et al., 2020; Wang et al., 2026).

The selected CVAE–GNO checkpoint gives the lowest observed endpoint error and neural inference time

Residual graph-kernel update

$$v_i^{\ell+1} = \sigma\left(W v_i^{\ell} + \sum_{j \in \mathcal{N}(i)} K_\theta(h_i, h_j, \mu) v_j^{\ell}\right),$$

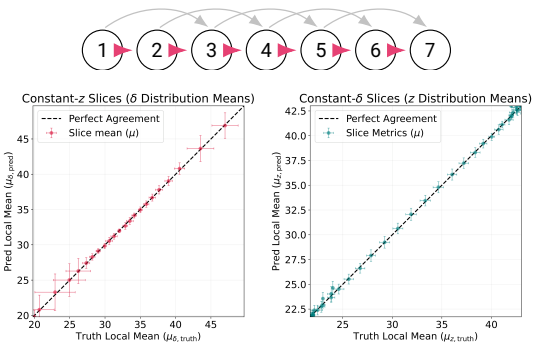
$$z_{i+1} = z_i + \Delta z_i.$$

The graph support is directed and local, the kernel is an effective latent interaction.

Selected checkpoint

$$\epsilon_{\mathrm{rel}} = 0.511, \quad \mathrm{SW} = 0.014,$$

$$\tau = 2.338 \text{ ms/sample}.$$



GNO conditional-slice consistency for the longitudinal marginal.

Sources: (Kovachki et al., 2023; Li et al., 2020).

Windowed attention and GNO differ mainly in kernel parameterization and graph support

Common discrete operator

$$(\mathcal{T}_\Theta h)_i = h_i + \sum_j A_{ij} K_\Theta(i, j, h_i, h_j, \mu) h_j.$$

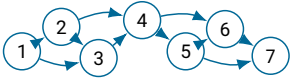
$$A_{ij}^{\mathrm{TF}} = \mathbf{1}_{0 \le i-j \le w}, \quad A_{ij}^{\mathrm{GNO}} = \mathbf{1}_{j \in \mathcal{N}(i)}.$$

1. Windowed attention and GNO formalism are equivalent architecturally, but the kernels are parameterized differently.
2. The difference lies mainly in the support given to each one of the kernels in the transformation.

Sources: (Li et al., 2020; Vaswani et al., 2017).



windowed attention: fixed band graph



GNO: message passing follows the machine graph

Depth composes local interactions

After L layers, local support can transmit information over multiple graph hops without direct all-to-all coupling.

Structured propagators reduce the observed endpoint error, but the numerical ranking is descriptive

Model	ϵ_{rel}	SW	Bias norm	τ [ms/sample]
CVAE only, no propagation	0.367	0.0096	–	–
Transformer, no window	0.681	0.087	17.337	18.74
Transformer, $w = 1$	0.650	0.026	52.389	14.35
Windowed Transformer ($w = 8$)	0.534	0.022	50.414	12.01
CVAE–GNO	0.511	0.014	0.376	2.338

Global model comparison

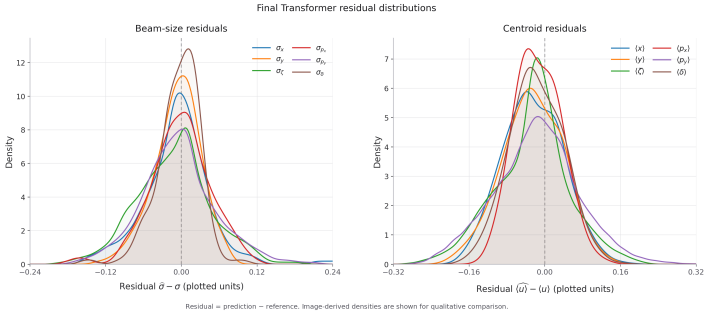
GNO present the lowest observed relative histogram and sliced-Wasserstein errors on the common held-out endpoint test set. Within the common neural benchmark, GNO is about eight times faster than global TF, under same hardware conditions.

Sources: Transport metric: (Peyré and Cuturi, 2019).

Error quantification

The CVAE–GNO error includes the pre-existing CVAE representation loss. Further uncertainty quantification is needed to test the accuracy and reliability of the learned propagation map.

A separate statistics head tests whether normalized shape can be returned to physical coordinates



Stacked residuals for the physical-statistics head, validation sample $s = 3$. The model predicts the first two moments of each physical axis.

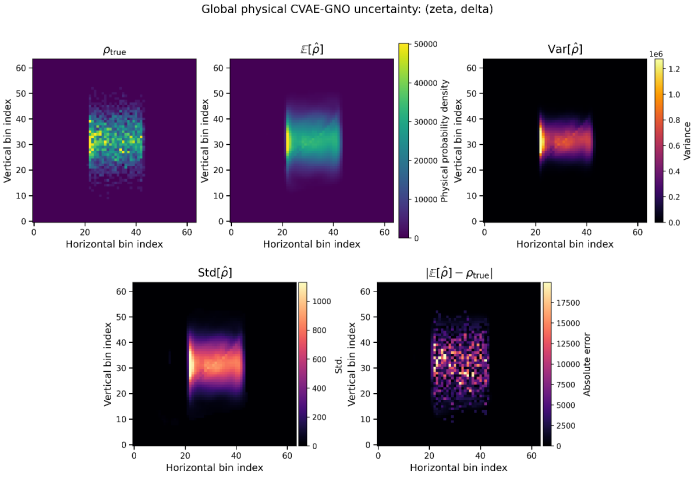
Physical head

Unit-mass histograms use sample-adaptive support. The model predicts

$$\hat{q}_N = (\widehat{m}_N, \widehat{\sigma}_N) \in \mathbb{R}^{12}$$

to recover physical axes and bin areas. Centroids test first moments, coordinate-wise RMS spreads test physical extension. Only σ_x and σ_y are conventional transverse beam sizes under the stated optics assumptions.

For one fixed sample, prediction bias is much larger than the conditional latent spread



Normalized mass

$$\frac{\text{mean}\,|\overline{H} - H_{\text{true}}|}{\text{mean}\,s_H} = 19.6.$$

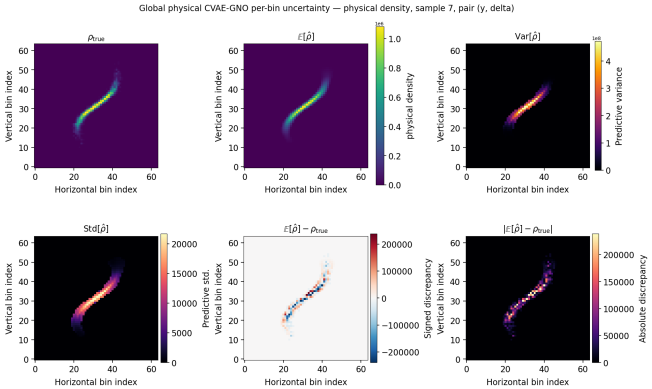
Physical density

$$\frac{\text{mean}\,|\overline{\rho} - \rho_{\text{true}}|}{\text{mean}\,s_\rho} = 55.5.$$

The larger ratio reflects inverse-bin-area rescaling, not an independent uncertainty source.

The ensemble is tightly clustered around a biased mean, increasing R does not remove the systematic discrepancy.

The held-out diagnostic remains under-dispersed even where absolute variance is largest



$R = 64$ stochastic passes for held-out sample $s = 7$, (y, δ) projection.

Global comparison

$$R_{\mathrm{global}} = \frac{3.487909 \times 10^3}{4.686791 \times 10^2} \simeq 7.44.$$

Highest-variance bin

The ensemble mean overpredicts the reference by about 10.7%, latent spread is about 2.38%, so the mean lies approximately $4.5\sigma_{\mathrm{latent}}$ from the reference.

Structured operator support is promising, while representation and uncertainty remain the main limits

Key messages from the 6D surrogate study

- Collective effects motivate surrogate modelling of distribution evolution.
- The FNO connects reduced longitudinal dynamics with spectral wake response.
- In the full-lattice endpoint task, local attention and graph-kernel propagation improve over dense causal mixing in the selected runs.

Conditions on those conclusions

- The model is bound by the sampled parameter domain,
- The 15 projections are not a unique 6D reconstruction,
- The latent ensemble is not calibrated predictive uncertainty,
- Intermediate dynamics and omitted physical effects require new simulation data.

Next step

A broader, checkpointed and uncertainty-calibrated FCC-ee HEB surrogate.

Sources: (Ghribi et al., 2026b).

The complete 6D result is traceable to public code, fixed data splits and archived machine inputs

Public outputs

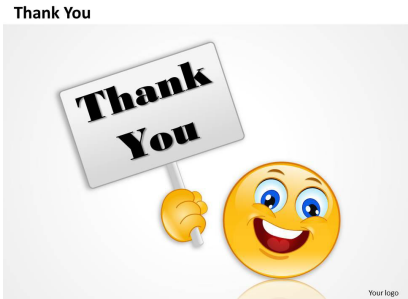
- ML4CollEffects GitHub repository,
- Zenodo dataset, DOI 10.5281/zenodo.21644178,
- IPAC'26 paper MOP1072.

Contribution beyond the preliminary paper

Full CVAE formulation and latent diagnostics, global/local Transformer comparison, graph-kernel interpretation, GNO benchmark, physical-statistics head, conditional latent uncertainty, under-dispersion diagnosis, explicit limitations and reproducibility record.

Sources: (Ghribi et al., 2026b; Martinez Saenz, 2026a,b).

Thank you





Questions?

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Fifteen pairwise marginals preserve useful structure but do not determine the full 6D joint density

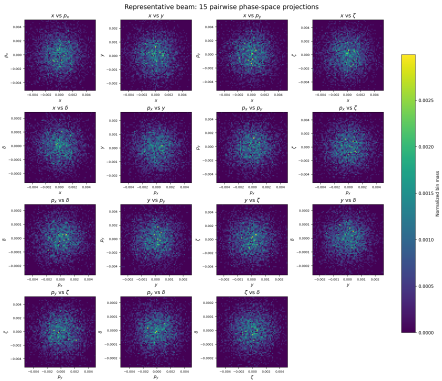


Fig. 1: fifteen two-dimensional marginals and the separate centroid/RMS vectors.

Sources: (Hoover and Wong, 2024).

Marginal representation

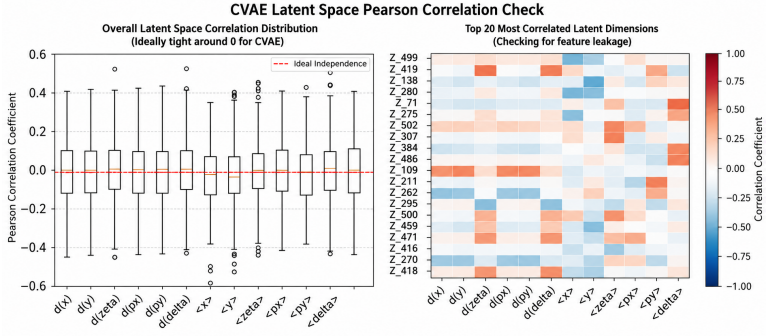
$$H(\rho) \in \mathbb{R}^{15\times 64\times 64}, \quad \sum_{a,b} H_{ij}[a,b] = 1.$$

Six centroids and six coordinate-wise RMS spreads restore physical position and scale.

Uniqueness of the 6D density

Three- to six-coordinate dependence is absent from the representation. The CVAE error is therefore an error on the fifteen projections, not on an unknown complete 6D density.

The latent coordinates show no pervasive linear alignment with the 12 explicit beam statistics



Sources: CVAE formulation: (Kingma and Welling, 2014; Sohn et al., 2015).

Test

$512 \times 12 = 6144$ Pearson coefficients on $N_{\text{val}} = 51$ samples.

Observed

Mean $|r| = 0.1161$, median $|r| = 0.1006$, maximum $|r| = 0.5333$.

In summary

Evidence against systematic linear leakage, not proof of independence or nonlinear disentanglement.

Windowed attention is a GNO with a normalized exponential kernel

Common causal support: $\mathcal{N}_i \equiv \mathcal{N}_{\overline{w}}(i) = \{j : \max(1, i-w+1) \le j \le i\}$. Score: $s_{ij}^{(\ell)} = q_i^{(\ell)\top} k_j^{(\ell)} / \sqrt{d_k}$.

1. Write local attention as a graph kernel

$$\alpha_{ij}^{(\ell,w)} = \mathbf{1}_{j\in\mathcal{N}_i} \frac{\exp(s_{ij}^{(\ell)})}{\sum_{r\in\mathcal{N}_i} \exp(s_{ir}^{(\ell)})}.$$

$$K_{\mathrm{TF,ij}}^{(\ell)} = \alpha_{ij}^{(\ell,w)} W_O^{(\ell)} W_V^{(\ell)},$$

$$\widetilde{h}_{i,\mathrm{TF}}^{(\ell+1)} = h_i^{(\ell)} + \sum_{j\in\mathcal{N}_i} K_{\mathrm{TF,ij}}^{(\ell)} h_j^{(\ell)}.$$

2. Select the corresponding GNO objects

$$\widetilde{h}_{i,\mathrm{GNO}}^{(\ell+1)} = h_i^{(\ell)} + \sum_{j\in\mathcal{N}_{\mathrm{GNO}}(i)} \omega_{ij} K_{\theta,ij}^{(\ell)} h_j^{(\ell)}.$$

$$\mathcal{N}_{\mathrm{GNO}}(i) = \mathcal{N}_i, \quad \omega_{ij} = 1,$$

$$K_{\theta,ij}^{(\ell)} = K_{\mathrm{TF,ij}}^{(\ell)} = \alpha_{ij}^{(\ell,w)} W_O^{(\ell)} W_V^{(\ell)}.$$

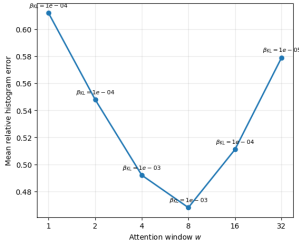
Exact consequence after substitution

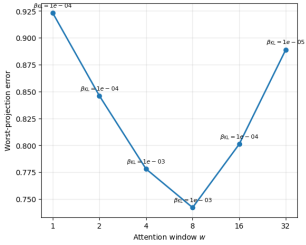
$$\widetilde{h}_{i,\mathrm{GNO}}^{(\ell+1)} = h_i^{(\ell)} + \sum_{j\in\mathcal{N}_i} K_{\mathrm{TF,ij}}^{(\ell)} h_j^{(\ell)} = \widetilde{h}_{i,\mathrm{TF}}^{(\ell+1)}.$$

Exact for one head; multi-head = kernel sum, while a learned GNO need not obey the softmax form.

Sources: (Li et al., 2020; Vaswani et al., 2017).

The single-seed window sweep has a non-monotonic minimum at intermediate support





Observed trend

- Very narrow support omits multi-element context,
- Intermediate support gives the lowest validation error in this run,
- Broad support progressively recovers dense upstream mixing.

w	β_{KL}	Mean error	Worst	Mean KL
1	10^{-4}	.612	.923	.418
2	10^{-4}	.548	.846	.356
4	10^{-3}	.492	.778	.238
8	10^{-3}	.468	.742	.211
16	10^{-4}	.511	.801	.304
32	10^{-5}	.579	.889	.447

Sources: Attention architecture: (Vaswani et al., 2017).

The reported variance measures sensitivity to CVAE posterior sampling at fixed input and fixed weights

Random quantity and Monte-Carlo estimator

$$z_0^{(r)} = \mu_\phi(X_0, c) + \sigma_\phi(X_0, c) \odot \epsilon^{(r)}, \quad \epsilon^{(r)} \sim \mathcal{N}(0, I),$$

$$Y^{(r)} = \mathcal{D}_\psi\left(\mathcal{T}_\theta^N(z_0^{(r)}, \{h_i\}, \mu), c_N\right),$$

$$\widehat{\mathrm{Var}}_z[Y \mid X_0, c, \mu, \theta] = \frac{1}{R}\sum_{r=1}^{R}\left(Y^{(r)} - \overline{Y}\right)^2.$$

Y may be one histogram bin, physical density bin, centroid or RMS spread.

Fixed inputs and weights

Input beam, conditioning variables, machine parameters, lattice tokens and trained weights θ.

Not included in total predictive uncertainty

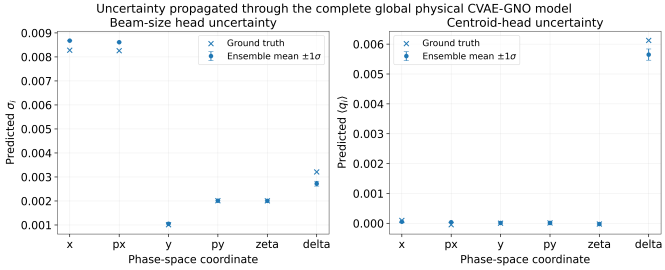
Weight/parameter uncertainty, finite-data effects, architecture and model-form discrepancy, simulator discrepancy, measurement noise and OOD error.

Measured statistics

Variability caused by the stochastic conditional latent state of this one trained CVAE–GNO model.

Sources: Variational posterior: (Kingma and Welling, 2014; Sohn et al., 2015).

Latent sampling does not cover the dominant energy-spread error of the physical-statistics head



Ensemble mean $\pm$ one latent-sampling standard deviation and reference values for six RMS spreads and centroids.

Dimensionless diagnostics

$$e_{\sigma,d} = \frac{\overline{\sigma}_d - \sigma_d^{\mathrm{true}}}{\sigma_d^{\mathrm{true}}}, \qquad b_{m,d} = \frac{\overline{m}_d - m_d^{\mathrm{true}}}{\sigma_d^{\mathrm{true}}}.$$

Selected sample

Transverse and longitudinal-position statistics are mostly within a few percent of the relevant beam scale. For δ, the predicted spread is only 11.8% of the reference value, while latent-sampling variability is about 0.026% of that reference spread.

The archived Xsuite dataset fixes the generation chain and train–validation–test split

Full-lattice chain

FCC-ee HEB MAD-X optics $\rightarrow$ cpymad $\rightarrow$ Xtrack 0.48.4 + PyWIT impedance/wake model $\rightarrow (Z_{\mathrm{in}}, \mu, Z_{\mathrm{out}})$.

$$\mathcal{D}_{\mathrm{track}} = \left\{ \left(Z_{\mathrm{in}}^{(s)}, \mu^{(s)}, Z_{\mathrm{out}}^{(s)} \right) \right\}_{s=1}^{2048}.$$

Only the upstream and final downstream clouds are stored.

Sources: (Martinez Saenz, 2026a,b).

Archive content

Fixed split indices, 4096 retained particles/cloud, data-generation and preprocessing scripts, environment, lattice source/checksum, parameter ranges/units, array manifest.

Each lattice token contains seven normalized physical parameters

$$\mu_i = (L_i, K_{1,i}, K_{2,i}, \phi_i, V_{\mathrm{rf},i}, f_{\mathrm{rf},i}, \varphi_{\mathrm{rf},i}) \in \mathbb{R}^7.$$

Quantity	Meaning	Pre-embedding scaling
L_i	element length	$L_i/(1\,\mathrm{m})$
$K_{1,i}$	quadrupole strength	$K_{1,i}/(10\,\mathrm{m}^{-2})$
$K_{2,i}$	sextupole strength	$K_{2,i}/(10\,\mathrm{m}^{-3})$
ϕ_i	bending angle	$\phi_i/(2\pi)$
$V_{\mathrm{rf},i}$	RF voltage	$\log(1 + V_{\mathrm{rf},i})$
$f_{\mathrm{rf},i}$	RF frequency	$\log(1 + f_{\mathrm{rf},i})$
$\varphi_{\mathrm{rf},i}$	RF phase	$\varphi_{\mathrm{rf},i}/(2\pi)$

Sources: (Wang et al., 2026).

The PyColleff task is a separate controlled longitudinal benchmark

Fixed configuration

Ring energy	3 GeV
Grid	$N_\zeta = 2048$
Resonator quality factor	$Q = 10^4$
Relaxation steps	$K = 3$
Under-relaxation	$\alpha = 0.7$
Samples	2048

Uniform conditioning ranges

I_b	[200,350] A
R	[5,12] mm
t_{NEG}	[1,3] μm

Sources: (de Sá and Alves, 2024).

Normalized mass becomes physical density only after coordinate scales define the bin areas

Shape representation

$$\sum_{a,b} H_{ij}^{(s)}[a,b] = 1.$$

The CVAE learns projected shape in normalized probability-mass coordinates.

Physical reconstruction

$$\rho_{ij}^{(s)}[a,b] = \frac{H_{ij}^{(s)}[a,b]}{\Delta\xi_{i,a}^{(s)}\Delta\xi_{j,b}^{(s)}}.$$

Bin edges are determined by the predicted centroids and RMS spreads.

Terminology

RMS spread remains defined for non-Gaussian distributions with finite second moment but does not characterize tails, asymmetry or multimodality. Only the transverse spatial spreads are conventional beam sizes under the stated optics assumptions.

The CVAE and endpoint propagators use different objectives

Snapshot CVAE pretraining

$$\mathcal{L}_{\mathrm{CVAE}} = \mathcal{L}_{\mathrm{rec}} + 10^{-2}\mathcal{L}_{\mathrm{KL}} + 10^{-3}\mathcal{L}_{\sigma} + 10^{-3}\mathcal{L}_{\mathrm{cent}}.$$

Endpoint propagation

$$\mathcal{L}_{\mathrm{prop}} = \mathcal{L}_{\mathrm{hist},N} + \beta_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + 10\mathcal{L}_{\log\sigma,N} + 10\mathcal{L}_{m,N}.$$

Training scope

The decoder is frozen during the window study, encoder/posterior heads, tokenizer and latent propagator remain trainable. Later, the distribution branch is frozen while the physical-statistics head is fine-tuned. No intermediate physical checkpoint appears in the loss.

Sources: (Kingma and Welling, 2014; Sohn et al., 2015).

The reported metrics probe complementary properties of the reconstructed distribution

Metric	Symbol	Interpretation
Relative histogram error	ϵ_{rel}	Projected mass discrepancy
Mean absolute centroid error	$\|\widehat{\mathbf{m}} - \mathbf{m}\|_1/6$	Centroid accuracy
Mean RMS-spread error	$\|\hat{\sigma} - \sigma\|_1/6$	Coordinate-scale accuracy
Wasserstein distance	$W_1(\hat{\rho}, \rho)$	Geometric mass displacement
Sliced Wasserstein distance	$\mathrm{SW}(\hat{\rho}, \rho)$	Distributional transport discrepancy
Residual bias	$\mathbb{E}[\widehat{H} - H]$	Systematic reconstruction structure
Inference time	t_{inf}	Neural forward-evaluation cost

$$\epsilon_{\mathrm{rel}} = \frac{1}{15B} \sum_{s=1}^{B} \sum_{p=1}^{15} \frac{\|\widehat{H}_{N,p}^{(s)} - H_{N,p}^{(s)}\|_1}{\|H_{N,p}^{(s)}\|_1 + \varepsilon}.$$

Sources: (Peyré and Cuturi, 2019).

The full window table confirms the same single-seed non-monotonic trend

w	β_{KL}	Selected epoch	Mean rel. error	Worst-proj. error	Mean KL
1	10^{-4}	18	0.612	0.923	0.418
2	10^{-4}	24	0.548	0.846	0.356
4	10^{-3}	31	0.492	0.778	0.238
8	10^{-3}	34	0.468	0.742	0.211
16	10^{-4}	27	0.511	0.801	0.304
32	10^{-5}	21	0.579	0.889	0.447

Scope

$w = 8$ is the lowest observed validation value in this run, no seed-to-seed error bars are available.

The complete fixed-sample uncertainty table separates latent spread from ensemble-mean discrepancy

Representation	Mean std.	Maximum std.	Mean variance	Maximum variance	Mean abs. discrepancy	Maximum abs. discrepancy
Normalized mass	3.748986×10^{-6}	3.138230×10^{-4}	6.816308×10^{-10}	9.848490×10^{-8}	7.364131×10^{-5}	1.159624×10^{-2}
Physical density	4.003429×10^{2}	3.244647×10^{4}	7.850748×10^{6}	1.052774×10^{9}	2.222052×10^{4}	3.394383×10^{6}

Interpretation

The ensemble varies the conditional latent state of one fixed trained model. It does not marginalize over weights or absorb measurement noise, finite-data uncertainty, architecture uncertainty, model discrepancy or OOD error.

The nominal Xsuite-to-GNO ratio is about , not a controlled simulator speed-up

Stage	Samples	Particles/grid	Time/sample	Total
Xsuite tracking	2048	10^6 particles	1.41 s	2904.52 s
PyColleff	2048	2048 points	2.72 s	5568.32 s
Histogram generation	2048	10^6 particles	2.32 ms	4.75 s

$$\frac{1.41\,\mathrm{s}}{2.338 \times 10^{-3}\,\mathrm{s}} \simeq 603.$$

Qualification

Simulator and neural timings were not measured under one hardware-normalized protocol, batching, hardware, precision, parallelization, cache state and software conditions differ.

The main beam-physics identities are not hard constraints in the present surrogate

Constraint/property	Status in this work
Exact symplecticity	not imposed on the learned 6D surrogate
Charge conservation	normalized histogram mass is enforced, no full-6D conservation law is imposed
Panofsky–Wenzel	motivates wake consistency, not a loss or architectural identity
Haïssinski equilibrium	motivates the longitudinal benchmark, not a full-lattice checkpoint constraint
Intermediate trajectory	not validated because simulator checkpoints are absent
Full 6D joint consistency	not guaranteed by 15 independently represented marginals
Calibrated predictive UQ	not achieved, the conditional latent ensemble is under-dispersed

Sources: (Huang et al., 2024; Métral, 2021; Zhou et al., 2023).

A reliable accelerator surrogate requires checkpoint supervision, broader 6D representations and calibrated uncertainty

Immediate validation

- Intermediate simulator checkpoints,
- Repeated seeds and deep ensembles,
- Broader parameter domains and OOD tests,
- Point-cloud/flow 6D comparisons,
- Physical constraints and conservation tests,
- Empirical coverage calibration.

Potential after validation

- Sensitivity analysis,
- Multi-parameter scans,
- Design-space reduction,
- Bayesian optimization under constraints,
- Simulation/measurement calibration,
- Uncertainty-aware digital-twin workflows.

Sources: (Hoover and Wong, 2024; Roussel et al., 2024).